

Reference angles and trigonometric functions



- 1 Consider the trigonometric ratio $\sin 135^\circ$.
 - a State which quadrant 135° is located.
 - b Find the measure of the positive acute angle that 135° is related to.
 - c Is $\sin 135^\circ$ positive or negative?
 - d Rewrite $\sin 135^\circ$ in terms of its reference angle.

- 2 Consider the trigonometric ratio $\cos 240^\circ$.
 - a State which quadrant 240° is located.
 - b Find the measure of the positive acute angle that 240° is related to.
 - c Is $\cos 240^\circ$ positive or negative?
 - d Rewrite $\cos 240^\circ$ in terms of its reference angle.

- 3 Consider the trigonometric ratio $\tan 210^\circ$.
 - a State which quadrant 210° is located.
 - b Find the measure of the positive acute angle that 210° is related to.
 - c Is $\tan 210^\circ$ positive or negative?
 - d Rewrite $\tan 210^\circ$ in terms of its reference angle.

- 4 Consider the trigonometric ratio $\sin \frac{7\pi}{6}$.
 - a State which quadrant $\frac{7\pi}{6}$ is located.
 - b Find the measure of the positive acute angle that $\frac{7\pi}{6}$ is related to.
 - c Is $\sin \frac{7\pi}{6}$ positive or negative?
 - d Rewrite $\sin \frac{7\pi}{6}$ in terms of its reference angle.

5 Consider the trigonometric ratio $\cos \frac{11\pi}{6}$.



- a State which quadrant $\frac{11\pi}{6}$ is located.
- b Find the measure of the positive acute angle that $\frac{11\pi}{6}$ is related to.
- c Is $\cos \frac{11\pi}{6}$ positive or negative?
- d Rewrite $\cos \frac{11\pi}{6}$ in terms of its reference angle.

6 Consider the trigonometric ratio $\tan \frac{5\pi}{4}$.

- a State which quadrant $\frac{5\pi}{4}$ is located.
- b Find the measure of the positive acute angle that $\frac{5\pi}{4}$ is related to.
- c Is $\tan \frac{5\pi}{4}$ positive or negative?
- d Rewrite $\tan \frac{5\pi}{4}$ in terms of its reference angle.

7 Consider the trigonometric ratio $\tan \frac{11\pi}{6}$.

- a State which quadrant $\frac{11\pi}{6}$ is located.
- b Find the measure of the positive acute angle that $\frac{11\pi}{6}$ is related to.
- c Is $\tan \frac{11\pi}{6}$ positive or negative?
- d Rewrite $\tan \frac{11\pi}{6}$ in terms of its reference angle.

8 Consider the trigonometric ratio $\cos \frac{7\pi}{4}$.

- a State which quadrant $\frac{7\pi}{4}$ is located.
- b Find the measure of the positive acute angle that $\frac{7\pi}{4}$ is related to.
- c Is $\cos \frac{7\pi}{4}$ positive or negative?
- d Rewrite $\cos \frac{7\pi}{4}$ in terms of its reference angle.

9 Consider the trigonometric ratio $\sin \frac{4\pi}{3}$.



- a State which quadrant $\frac{4\pi}{3}$ is located.
- b Find the measure of the positive acute angle that $\frac{4\pi}{3}$ is related to.
- c Is $\sin \frac{4\pi}{3}$ positive or negative?
- d Which of the following is $\sin \frac{4\pi}{3}$ equivalent to: $-\sin \frac{\pi}{3}$ or $\sin \frac{\pi}{3}$?

10 Consider the trigonometric ratio $\cos \frac{4\pi}{3}$.

- a State which quadrant $\frac{4\pi}{3}$ is located.
- b Find the measure of the positive acute angle that $\frac{4\pi}{3}$ is related to.
- c Is $\cos \frac{4\pi}{3}$ positive or negative?
- d Which of the following is $\cos \frac{4\pi}{3}$ equivalent to: $-\cos \frac{\pi}{3}$ or $\cos \frac{\pi}{3}$?

11 Consider the trigonometric ratio $\tan \frac{7\pi}{4}$.

- a State which quadrant $\frac{7\pi}{4}$ is located.
- b Find the measure of the positive acute angle that $\frac{7\pi}{4}$ is related to.
- c Is $\tan \frac{7\pi}{4}$ positive or negative?
- d Which of the following is $\tan \frac{7\pi}{4}$ equivalent to: $\tan \frac{\pi}{4}$ or $-\tan \frac{\pi}{4}$?



Exact values of trigonometric functions

12 Consider the angle 810° .

- a** State the coordinates of the point on the unit circle that corresponds to 810° .
- b** Find the value of $\cos 810^\circ$.
- c** Find the value of $\sin 810^\circ$.
- d** Determine whether the following are undefined:
 - i** $\cot 810^\circ$
 - ii** $\csc 810^\circ$
 - iii** $\sec 810^\circ$
 - iv** $\tan 810^\circ$

13 Consider the angle (-360°) .

- a** State the coordinates of the point on the unit circle that corresponds to (-360°) .
- b** Find the value of $\cos(-360^\circ)$.
- c** Find the value of $\sin(-360^\circ)$.
- d** Determine whether the following are undefined:
 - i** $\cot(-360^\circ)$
 - ii** $\csc(-360^\circ)$
 - iii** $\sec(-360^\circ)$
 - iv** $\tan(-360^\circ)$

14 Consider the angle 270° .

- a** State the coordinates of the point on the unit circle that corresponds to 270° .
- b** Find the value of $\cos 270^\circ$.
- c** Find the value of $\sin 270^\circ$.
- d** Determine whether the following are undefined:
 - i** $\tan 270^\circ$
 - ii** $\cot 270^\circ$
 - iii** $\csc 270^\circ$
 - iv** $\sec 270^\circ$

15 Consider the angle (-180°) .

- a** State the coordinates of the point on the unit circle that corresponds to (-180°) .
- b** Find the value of $\cos(-180^\circ)$.
- c** Find the value of $\sin(-180^\circ)$.
- d** Determine whether the following are undefined:
 - i** $\sec(-180^\circ)$
 - ii** $\cot(-180^\circ)$
 - iii** $\csc(-180^\circ)$
 - iv** $\tan(-180^\circ)$

16 Consider the angle 720° .



- a State the coordinates of the point on the unit circle that corresponds to 720° .
- b Find the value of $\cos 720^\circ$.
- c Find the value of $\sin 720^\circ$.
- d Determine whether the following are undefined:

i $\sec 720^\circ$ ii $\cot 720^\circ$ iii $\tan 720^\circ$ iv $\csc 720^\circ$

17 Consider the angle 1260° .

- a State the coordinates of the point on the unit circle that corresponds to 1260° .
- b Find the value of $\cos 1260^\circ$.
- c Find the value of $\sin 1260^\circ$.
- d Determine whether the following are undefined:

i $\csc 1260^\circ$ ii $\cot 1260^\circ$ iii $\tan 1260^\circ$ iv $\sec 1260^\circ$

18 Find the exact value of the following trigonometric ratios:

- | | | | |
|--------------------|--------------------|--------------------|--------------------|
| a $\sin 330^\circ$ | b $\cos 330^\circ$ | c $\tan 330^\circ$ | d $\csc 330^\circ$ |
| e $\sec 330^\circ$ | f $\cot 330^\circ$ | g $\sin 510^\circ$ | h $\cos 510^\circ$ |
| i $\tan 510^\circ$ | j $\csc 510^\circ$ | k $\sec 510^\circ$ | l $\cot 510^\circ$ |

19 Find the exact value of the following trigonometric ratios:

- | | | | |
|-------------------------|-------------------------|-------------------------|-------------------------|
| a $\sin \frac{2\pi}{3}$ | b $\cos \frac{2\pi}{3}$ | c $\tan \frac{2\pi}{3}$ | d $\csc \frac{2\pi}{3}$ |
| e $\sec \frac{2\pi}{3}$ | f $\cot \frac{2\pi}{3}$ | | |

20 For each of the following trigonometric ratios:

- i Find the value of the reference angle.
- ii Find the exact value of the trigonometric ratio.

- | | | |
|----------------------|---------------------|--------------------------------------|
| a $\cos(-240^\circ)$ | b $\sin 5400^\circ$ | c $\cos\left(-\frac{4\pi}{3}\right)$ |
|----------------------|---------------------|--------------------------------------|

21 Find the exact values of the following:



a $\cos 0^\circ$

b $\sin 90^\circ$

c $\cos(-180^\circ)$

d $\sin 630^\circ$

e $\tan 90^\circ$

f $\tan(-630^\circ)$

g $\cot 30^\circ$

h $\csc 60^\circ$

i $\sec 30^\circ$

j $\sec 45^\circ$

k $\csc 45^\circ$

l $\cot 45^\circ$

m $\cot 270^\circ$

n $\sec 180^\circ$

o $\csc 0^\circ$

p $\cot 450^\circ$

q $\csc 900^\circ$

r $\sec(-450)^\circ$

22 Find the exact values of the following:

a $\sin \frac{\pi}{2}$

b $\sin\left(-\frac{\pi}{6}\right)$

c $\cos\left(-\frac{5\pi}{3}\right)$

d $\cos 4\pi$

e $\sin \frac{5\pi}{6}$

f $\cos \frac{3\pi}{4}$

g $\sin \frac{7\pi}{6}$

h $\cos \frac{7\pi}{6}$

i $\tan \frac{7\pi}{6}$

j $\sin \frac{5\pi}{3}$

k $\cos \frac{5\pi}{3}$

l $\tan \frac{3\pi}{4}$

m $\tan \frac{11\pi}{6}$

n $\tan \frac{3\pi}{2}$

o $\sec \frac{\pi}{2}$

p $\cot 3\pi$

23 By rewriting each ratio in terms of the related acute angle, determine the exact value of the following:

a
$$\frac{\sin 120^\circ \cos 240^\circ \tan 330^\circ}{\tan(-45)^\circ}$$

b
$$\frac{\left(\sin \frac{2\pi}{3}\right) \left(\cos \frac{2\pi}{3}\right) \left(\tan \frac{3\pi}{4}\right)}{\tan\left(-\frac{\pi}{4}\right)}$$

c
$$\frac{\sin\left(\frac{2\pi}{3}\right) + \cos\left(\frac{5\pi}{6}\right) - \tan\left(\frac{7\pi}{4}\right)}{\cos\left(\frac{4\pi}{3}\right)}$$

d
$$\frac{-\sin\left(\frac{5\pi}{3}\right) + \cos\left(\frac{7\pi}{6}\right) + \tan\left(\frac{5\pi}{3}\right)}{-\cos\left(\frac{2\pi}{3}\right)}$$